

Salesforce:16

CC- Student Record Management System Develop an application in Salesforce using the Apex programming language and visual force page to manage student records (Name, Roll No, Class, Mobile No)stored in a custom object.

1 Create a custom object

2 Create Tab for Student Object

Go to:

◆ **Setup → Search Tabs**

Click:

👉 **Object Manager Tabs**

Then:

1. Click **New**
2. Select:

Student

3. Choose any icon 🎯
4. Click **Next → Next → Save**

VFS PAGE:

```
<apex:page controller="StudentController1">
```

```
<apex:form>
```

```
<apex:pageBlock title="Student Record Management">
```

```
<apex:pageMessages />
```

```
<apex:pageBlockSection columns="2">
```

```
<apex:inputText value="{!student.Name}" label="Name" required="true"/>
```

```
<apex:inputText value="{!student.Roll_No__c}" label="Roll No" required="true"/>
```

```
<apex:inputText value="{!student.Class__c}" label="Class"/>
```

```
<apex:inputText value="{!student.Mobile_No__c}" label="Mobile No"/>
```

```
<apex:inputText value="{!student.Email__c}" label="Email"/>
```

```
</apex:pageBlockSection>
```

```
<apex:pageBlockButtons>
```

```
    <apex:commandButton value="Save" action="{!saveStudent}" rerender="tableBlock,msg"/>
```

```
    <apex:commandButton value="Update" action="{!updateStudent}"  
rerender="tableBlock,msg"/>
```

```
</apex:pageBlockButtons>
```

```
</apex:pageBlock>
```

```
<!-- Table -->
```

```
<apex:pageBlock title="Student List" id="tableBlock">
```

```
    <apex:pageBlockTable value="{!studentList}" var="s">
```

```
        <apex:column value="{!s.Name}" headerValue="Name"/>
```

```
        <apex:column value="{!s.Roll_No__c}" headerValue="Roll No"/>
```

```
        <apex:column value="{!s.Class__c}" headerValue="Class"/>
```

```
        <apex:column value="{!s.Mobile_No__c}" headerValue="Mobile"/>
```

```
        <apex:column value="{!s.Email__c}" headerValue="Email"/>
```

```
        <apex:column headerValue="Actions">
```

```
            <apex:commandLink value="Edit" action="{!editStudent}" rerender="tableBlock,msg">
```

```
                <apex:param value="{!s.Id}" assignTo="{!student.Id}"/>
```

```
            </apex:commandLink>
```

```
        &nbsp; | &nbsp;
```

```
        <apex:commandLink value="Delete" action="{!deleteStudent}"
rerender="tableBlock,msg">
```

```
        <apex:param value="{!s.Id}" assignTo="{!student.Id}"/>
```

```
    </apex:commandLink>
```

```
</apex:column>
```

```
</apex:pageBlockTable>
```

```
</apex:pageBlock>
```

```
</apex:form>
```

```
</apex:page>
```

Controller:

```
public with sharing class StudentController1 {
```

```
    public Student__c student { get; set; }
```

```
    public List<Student__c> studentList { get; set; }
```

```
    public StudentController1() {
```

```
        student = new Student__c();
```

```
        loadStudents();
```

```
    }
```

```
// Load all students
```

```
public void loadStudents() {
```

```
    studentList = [
```

```
        SELECT Id, Name, Roll_No__c, Class__c, Mobile_No__c, Email__c
```

```

        FROM Student__c

        ORDER BY CreatedDate DESC

    ];
}

// Email validation
public Boolean isValidEmail(String email) {
    if (String.isBlank(email)) return false;

    Pattern p = Pattern.compile('^([A-Za-z0-9+_-]+@[A-Za-z0-9.-]+)$');
    Matcher m = p.matcher(email);
    return m.matches();
}

// SAVE
public void saveStudent() {

    // Name validation
    if (String.isBlank(student.Name) || student.Name.length() < 3) {
        ApexPages.addMessage(new ApexPages.Message(
            ApexPages.Severity.ERROR, 'Name must be at least 3 characters'
        ));
        return;
    }

    // Roll No validation
    if (student.Roll_No__c == null || student.Roll_No__c <= 0) {
        ApexPages.addMessage(new ApexPages.Message(
            ApexPages.Severity.ERROR, 'Roll No must be greater than 0'
        ));
        return;
    }
}

```

```
}
```

```
// Unique Roll No
```

```
List<Student__c> existing = [
```

```
    SELECT Id FROM Student__c
```

```
    WHERE Roll_No__c = :student.Roll_No__c
```

```
    LIMIT 1
```

```
];
```

```
if (!existing.isEmpty()) {
```

```
    ApexPages.addMessage(new ApexPages.Message(
```

```
        ApexPages.Severity.ERROR, 'Roll No must be unique'
```

```
    ));
```

```
    return;
```

```
}
```

```
// Email validation
```

```
if (!isValidEmail(student.Email__c)) {
```

```
    ApexPages.addMessage(new ApexPages.Message(
```

```
        ApexPages.Severity.ERROR, 'Invalid Email Format'
```

```
    ));
```

```
    return;
```

```
}
```

```
insert student;
```

```
ApexPages.addMessage(new ApexPages.Message(
```

```
    ApexPages.Severity.CONFIRM, 'Student Added Successfully'
```

```
));
```

```
student = new Student__c();
```

```

        loadStudents();
    }

// EDIT
public void editStudent() {
    student = [
        SELECT Id, Name, Roll_No__c, Class__c, Mobile_No__c, Email__c
        FROM Student__c
        WHERE Id = :student.Id
    ];
}

// UPDATE
public void updateStudent() {
    update student;

    ApexPages.addMessage(new ApexPages.Message(
        ApexPages.Severity.CONFIRM, 'Student Updated Successfully'
    ));

    student = new Student__c();
    loadStudents();
}

// DELETE
public void deleteStudent() {
    delete [SELECT Id FROM Student__c WHERE Id = :student.Id];
    loadStudents();
}
}

```